UNIVERSITY OF NAIROBI — FACULTY OF SCIENCE & TECHNOLOGY
DEPARTMENT OF COMPUTING & INFORMATICS | CHIROMO CAMPUS

UON CHIROMO YEAR 2: MASTER PAST EXAM PAPERS SOLVED COMPENDIUM (2018–2026)

Exhaustive Step-by-Step Model Marking Scheme Solutions, Mathematical Derivations & Algorithmic Proofs Across All Year 2 Examinations

Curriculum Year & Scope:	**Bachelor of Science in Computer Science — Year 2 (Full Scope)**
Academic Units Covered:	• SCS 3207: Operating Systems • SCS 3203: Computer Architecture • SCS 3205: Knowledge-Based Systems & Programming • SMA 2114: Differential & Integral Calculus • CSC 3211: Computer Networks
Examinations Solved:	UoN Semester Examinations & CATs (2018, 2019, 2024, 2025, 2026)
Primary Prescribed Texts:	• Tanenbaum (OS 4th ed.) | Stallings (OS 9th ed.) • Stallings (Comp Arch 10th ed.) | Patterson & Hennessy (MIPS) • Russell & Norvig (AIMA 3rd ed.) | Akerkar (KBS) • Forouzan (Networks 4th ed.) | Comer (Networks 6th ed.) • Kendall & Kendall (SAD 10th ed.) | Sommerville (SE 9th/10th ed.)
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Offline Readiness:	**100% SELF-CONTAINED — NO INTERNET CONNECTION REQUIRED**

PART 1: SCS 3207 — OPERATING SYSTEMS (SOLVED EXAMS & CATS)

Examination: UoN SCS 3207 Operating Systems — January 26, 2026

Question 1 (a) [6 Marks]: Explain the dual-mode operation of modern CPUs (User Mode vs. Kernel Mode). Illustrate how mode switching occurs during a system call with the hardware Mode Bit.

Model Marking Scheme Solution:
1. Architectural Purpose: Dual-mode operation prevents rogue user-level programs from directly executing privileged I/O instructions, halting the CPU, or corrupting physical memory allocated to other processes or the kernel.
2. Privilege Levels & Mode Bit:
• **User Mode (Mode Bit = 1):** The CPU executes unprivileged instructions. Direct access to physical hardware, MMU page tables, and I/O ports is prohibited.
• **Kernel / Supervisor Mode (Mode Bit = 0):** The CPU has unrestricted access to the entire memory space and all privileged instructions (e.g., `CLI`, `STI`, `LIDT`, `IN/OUT`).
3. Mode Transition Sequence (Step-by-Step):
1. User process invokes a library wrapper (e.g., `read()` in POSIX).
2. Wrapper places the system call number in register (e.g., `EAX`/`RAX` on x86) and parameters in registers.
3. The software trap/interrupt instruction (`SYSCALL` or `INT 0x80`) is executed.
4. **Hardware Action:** Hardware automatically saves User PC and Processor Status Word (PSW), switches the **Mode Bit from 1 to 0**, and jumps to the System Call Dispatcher address via the Interrupt Descriptor Table (IDT).
5. Kernel executes service routine, checks parameters for safety, writes return code to register.
6. Kernel executes `SYSRET` or `IRET`: hardware restores User PSW, flips **Mode Bit back from 0 to 1**, and resumes user execution at next instruction.

Question 3 (a) [15 Marks]: A counting semaphore S has an initial value of -2, and two processes P2 and P3 are in its waiting queue where P2 is ahead of P3. Another process P1 executes signal(S) (or V(S)), followed by process P4 executing wait(S) (or P(S)).
(i) Trace the value of semaphore S after each operation.
(ii) What is the state of the waiting queue at each step?
(iii) Which process is unblocked, and when?

Step #	Action Executed	Value of Semaphore S	Waiting Queue (FIFO)	Process State Transition & Explanation
Initial	System State	S = -2	[P2, P3]	Negative count indicates 2 processes waiting. P2 is head of queue.
Step 1	P1 executes signal(S)	S = -2 + 1 = -1	[P3]	S <= 0 triggers unblocking of head process. **P2 is unblocked** and moved to Ready queue. P3 remains waiting.
Step 2	P4 executes wait(S)	S = -1 - 1 = -2	[P3, P4]	S < 0 causes calling process P4 to block. **P4 is placed at tail of waiting queue** behind P3.

Question 2 (b) [10 Marks]: Consider 5 processes P0 through P4 and 3 resource types A (10 instances), B (5 instances), C (7 instances). Current allocation is P0(0,1,0), P1(2,0,0), P2(3,0,2), P3(2,1,1), P4(0,0,2). Max demand is P0(7,5,3), P1(3,2,2), P2(9,0,2), P3(2,2,2), P4(4,3,3). Determine if the system is in a safe state using Banker's Algorithm.

Model Marking Scheme Solution:
1. Calculate Total Allocated:
Total A = 0 + 2 + 3 + 2 + 0 = 7. Total B = 1 + 0 + 0 + 1 + 0 = 2. Total C = 0 + 0 + 2 + 1 + 2 = 5.
2. Calculate Available Vector:
Available = Total Instances - Total Allocated = (10 - 7, 5 - 2, 7 - 5) = **(3, 3, 2)**.
3. Calculate Need Matrix (Need = Max - Allocation):
• Need(P0) = (7,5,3) - (0,1,0) = (7, 4, 3)
• Need(P1) = (3,2,2) - (2,0,0) = (1, 2, 2)
• Need(P2) = (9,0,2) - (3,0,2) = (6, 0, 0)
• Need(P3) = (2,2,2) - (2,1,1) = (0, 1, 1)
• Need(P4) = (4,3,3) - (0,0,2) = (4, 3, 1)
4. Safety Algorithm Execution Trace:
• Work = (3, 3, 2). Check P0: Need (7,4,3) <= Work? No. (Wait)
• Check P1: Need (1,2,2) <= Work (3,3,2)? **YES!** P1 completes. Work = (3,3,2) + Allocation(P1)(2,0,0) = **(5, 3, 2)**.
• Check P3: Need (0,1,1) <= Work (5,3,2)? **YES!** P3 completes. Work = (5,3,2) + Allocation(P3)(2,1,1) = **(7, 4, 3)**.
• Check P4: Need (4,3,1) <= Work (7,4,3)? **YES!** P4 completes. Work = (7,4,3) + Allocation(P4)(0,0,2) = **(7, 4, 5)**.
• Check P0: Need (7,4,3) <= Work (7,4,5)? **YES!** P0 completes. Work = (7,4,5) + Allocation(P0)(0,1,0) = **(7, 5, 5)**.
• Check P2: Need (6,0,0) <= Work (7,5,5)? **YES!** P2 completes. Work = (7,5,5) + Allocation(P2)(3,0,2) = **(10, 5, 7)**.
Verdict: All processes finish. The system is in a **SAFE STATE**. Safe sequence: **<P1, P3, P4, P0, P2>**.

PART 2: SCS 3203 — COMPUTER ARCHITECTURE (SOLVED EXAMS & CATS)

Examination: UoN SCS 3203 Computer Architecture — January 20, 2026

Question 1 (b) [6 Marks]: Describe the three cache mapping techniques (Direct, Fully Associative, and Set Associative). Compare them in terms of hardware complexity and cache miss rate.

Mapping Technique	Address Bit Division	Placement Rule	Miss Rate (Conflict)	Hardware Complexity & Search
Direct Mapped	Tag | Line Index | Word Offset	Block mapped to exactly one cache line: line = (Block #) mod (Total Lines)	Highest (frequent conflict misses when multiple blocks compete for same line)	Lowest (Single comparator; instantaneous look-up).
Fully Associative	Tag | Word Offset	Block can be placed anywhere in cache.	Lowest (Zero conflict misses; only compulsory & capacity misses)	Highest (Requires parallel comparators for every single cache line; high silicon cost).
Set Associative (k-way)	Tag | Set Index | Word Offset	Cache divided into S sets of k lines. Block maps to set: set = (Block #) mod S	Balanced (drastically lowers conflict misses over direct mapping)	Moderate (Requires k comparators operating in parallel; industry standard).

Question 2 (a) [8 Marks]: Using Booth's Algorithm, compute the product of Multiplicand M = -5 (1011 in 4-bit 2's complement) and Multiplier Q = +3 (0011 in 4-bit 2's complement). Show the step-by-step register contents of A, Q, Q-1.

Model Marking Scheme Solution:
Multiplicand M = 1011 (-5). Negative of M (-M) = 0101 (+5). Multiplier Q = 0011 (+3).
Registers: A = 0000, Q = 0011, Q-1 = 0. Count = 4.

Cycle #	Action Taken	Register A	Register Q	Q-1	Count Remaining
Init	Initialize registers	0000	0011	0	4
Cycle 1	Q0 Q-1 = 10 -> A = A - M (A + 0101 = 0101) Arithmetic Shift Right (ASR)	0101 **0010**	0011 **1001**	0 **1**	3
Cycle 2	Q0 Q-1 = 11 -> No arithmetic op Arithmetic Shift Right (ASR)	0010 **0001**	1001 **1100**	1 **1**	2
Cycle 3	Q0 Q-1 = 01 -> A = A + M (A + 1011 = 1100) Arithmetic Shift Right (ASR)	1100 **1110**	1100 **0110**	1 **0**	1
Cycle 4	Q0 Q-1 = 00 -> No arithmetic op Arithmetic Shift Right (ASR)	1110 **1111**	0110 **0011**	0 **0**	0 (Done)

Result Concatenation: A:Q = 1111 0001.
In 8-bit 2's complement: 2's complement of 1111 0001 is (0000 1110 + 1) = 0000 1111 = 15. Hence, 1111 0001 = **-15**.
Proof: (-5) * (+3) = -15. **CORRECT.**

PART 3: SCS 3205 — KNOWLEDGE-BASED SYSTEMS & PROGRAMMING

Examination: UoN SCS 3205 Knowledge-Based Systems — January 22, 2026

Question 1 (c) [8 Marks]: Transform the following First-Order Logic (FOL) statement into Conjunctive Normal Form (CNF) clauses: *'Every student who studies hard passes their exams or is lucky.'* Show all standard conversion steps.

Model Marking Scheme Solution:
Step 1 (Formalize in FOL):
$\forall x [(Student(x) \land StudiesHard(x)) \rightarrow (PassesExam(x) \lor Lucky(x))]$
Step 2 (Eliminate Implication $P \rightarrow Q \equiv \neg P \lor Q$):
$\forall x [\neg(Student(x) \land StudiesHard(x)) \lor (PassesExam(x) \lor Lucky(x))]$
Step 3 (Apply De Morgan's Law to push negation inward):
$\forall x [\neg Student(x) \lor \neg StudiesHard(x) \lor PassesExam(x) \lor Lucky(x)]$
Step 4 (Standardize Variables): Variable x is already unique.
Step 5 (Skolemization): No existential quantifiers ($\exists$) exist. No Skolem functions needed.
Step 6 (Drop Universal Quantifiers):
$\neg Student(x) \lor \neg StudiesHard(x) \lor PassesExam(x) \lor Lucky(x)$
Final CNF Clause: { ¬Student(x), ¬StudiesHard(x), PassesExam(x), Lucky(x) } (Single disjunctive clause).

Question 2 (a) [6 Marks]: Explain the execution semantics of the Prolog cut operator (!). Give an example illustrating green cut vs. red cut.

Model Marking Scheme Solution:
1. Semantics of Cut (!): The cut operator commits Prolog to all choices made since the parent goal was unified with the clause head. It prunes all alternative choice points for goals preceding the cut and prevents backtracking into subsequent alternative clauses for the same predicate.
2. Green Cut (Safe / Efficiency):
• Prunes redundant search branches without altering the declarative meaning of the program.
• Example: max(X, Y, X) :- X >= Y, !. max(X, Y, Y) :- X < Y. (The cut purely prevents checking X < Y when X >= Y is already proven).
3. Red Cut (Hazardous / Procedural):
• Alters the logical declarative semantics. Removing the cut changes the truth set of solutions.
• Example: max(X, Y, X) :- X >= Y, !. max(X, Y, Y). (Omits explicit check in second clause. If cut is removed, query max(5, 2, 2) incorrectly succeeds!).

PART 4: SMA 2114 — DIFFERENTIAL & INTEGRAL CALCULUS (SOLVED EXAMS)

Examination: UoN SMA 2114 / SCS 3114 Calculus — 2025 Semester Exam

Question 1 (a) [6 Marks]: Evaluate the integral: $\int x^2 \cdot e^{(3x)}\, dx$ using the technique of Integration by Parts.

Model Marking Scheme Solution:
Formula: $\int u\, dv = u\, v - \int v\, du$.
First Application:
Let $u = x^2 \rightarrow du = 2x\, dx$. Let $dv = e^{(3x)}\, dx \rightarrow v = (1/3)\, e^{(3x)}$.
$\int x^2\, e^{(3x)}\, dx = (1/3)\, x^2\, e^{(3x)} - \int (2/3)\, x\, e^{(3x)}\, dx$.
Second Application on $\int x\, e^{(3x)}\, dx$:
Let $u = x \rightarrow du = dx$. Let $dv = e^{(3x)}\, dx \rightarrow v = (1/3)\, e^{(3x)}$.
$\int x\, e^{(3x)}\, dx = (1/3)\, x\, e^{(3x)} - (1/3) \int e^{(3x)}\, dx = (1/3)\, x\, e^{(3x)} - (1/9)\, e^{(3x)}$.
Substitute Back:
$= (1/3)\, x^2\, e^{(3x)} - (2/3) [(1/3)\, x\, e^{(3x)} - (1/9)\, e^{(3x)}] + C$
$=$ **$(1/3)\, x^2\, e^{(3x)} - (2/9)\, x\, e^{(3x)} + (2/27)\, e^{(3x)} + C$** $=$ **$(e^{(3x)} / 27) [9x^2 - 6x + 2] + C$**.

Question 2 (b) [8 Marks]: Solve the first-order linear differential equation: $dy/dx + (2/x)\, y = 4x$, with initial condition $y(1) = 2$.

Model Marking Scheme Solution:
Standard form: $dy/dx + P(x)\, y = Q(x)$, where $P(x) = 2/x$, $Q(x) = 4x$.
1. Integrating Factor I(x):
$I(x) = e^{(\int P(x)\, dx)} = e^{(\int (2/x)\, dx)} = e^{(2 \ln|x|)} = e^{(\ln(x^2))} =$ **x^2**.
2. Multiply Differential Equation by I(x):
$x^2\, (dy/dx) + 2x\, y = 4x^3 \rightarrow d/dx [x^2\, y] = 4x^3$.
3. Integrate Both Sides:
$x^2\, y = \int 4x^3\, dx = x$■ $+ C \rightarrow$ **$y(x) = x^2 + C / x^2$**.
4. Apply Initial Condition y(1) = 2:
$2 = 1^2 + C / 1^2 \rightarrow 2 = 1 + C \rightarrow$ **$C = 1$**.
Particular Solution: $y(x) = x^2 + 1/x^2$.